

Single-Electron-Sensitive SiC Detector for Energy Spectroscopy, Time-Resolved Characterization, and Scattering Electron Imaging in Electron Microscopy

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Abstract—This work demonstrates a single-electron-sensitive SiC detector featuring an iso-potential guard ring and low-leakage processing techniques, achieving 0.05 pA dark current at the bias of 200 V with 200°C stability, linear response covering a wide beam current range from 0.78 pA to 410 nA with a contrast ratio exceeding 10^4 over dark current, and a 95% electron detection efficiency. The detector demonstrates for the first time single-electron-detection-based temporal analysis on compositional and spatial information, where 92.5 ps time jitter in the time-to-digital converter and 6.62 ns time resolution in the detection system has been achieved, facilitating direct single electron energy spectroscopy. The detector superior ruggedness against 1.86 GeV heavy ions and 10^7 A/cm² 30 keV electron beam irradiation further verifies that, this single electron detector could elevate the electron microscopy from statistical averaging not only to single-quantum-event studies, but also to operations in harsh environment.

I. INTRODUCTION

Electron microscopy capable of simultaneously resolving compositional and topographic information has emerged as a critical inspection technique for state-of-the-art integrated circuits (ICs), which increasingly feature intricate structures with scaling-down dimensions and high aspect ratios [1-3]. However, a fundamental tradeoff persists between achieving high precision, requiring large electron doses for optimal signal-to-noise ratios, and minimizing electron bombardment damage to sensitive structures like thin gate dielectrics, which necessitates low-dose operation [2,4]. Highly sensitive electron detection, ideally reaching single-electron sensitivity, has thus become an essential requirement for next-generation microscopy, demanding exceptional performance from both detectors and electronics [2,5]. Effective leakage suppression in detector is critical to minimize dark current, while low parasitic capacitance and robust radiation tolerance are equally vital to ensure rugged electron detection and charge collection. Meanwhile, the front-end electronics should have high gain-bandwidth product and low noise to enable precise decoding of compositional and morphological information from the electron energy spectrum, time of arrival, and scattering angle.

In this work, we demonstrate a novel single electron detector based on wide-bandgap silicon carbide (4H-SiC). Leveraging its inherently low thermal carrier generation rate, advantageous temperature stability, and superior radiation hardness [6,7], the detector achieves sub-pA level dark current with temperature up to 200°C, enabling linear response to electron beams in a wide dynamic range. Scattering electron imaging with elevated shadow and thus three-dimensional effect is realized. And, for the first time, single electron detection of characteristic scattering electrons from different elements have been achieved, enabling real-time electron microscopy at single electrons level with compositional and topographic information directly decoded.

II. DEVICE FABRICATION

The 4H-SiC electron detector featuring the iso-potential guard ring and bevel mesa termination is illustrated in Fig. 1a. The p-type guard ring shares the same potential with the inner p-region, so as to shield the lateral electric field that would otherwise collect noise charge from the outside region. The epitaxial structure grown on a 6-inch 4°-off-axis n-type 4H-SiC substrate, consists of a 100 μ m n-absorbing layer ($N_D=1.85\times10^{14}$ cm⁻³), a 0.2 μ m p layer ($N_A=1\times10^{18}$ cm⁻³), and a 0.1 μ m p+ contact layer ($N_A=1\times10^{19}$ cm⁻³). As shown in Fig. 1b, the 100- μ m thick 4H-SiC epilayer for large detection efficiency and low capacitance has near-ideal lattice quality. The fabrication process starts with the 3° bevel-mesa termination fabrication with photoresist reflow and inductively coupled plasma etching with CF₄, obtaining an active region with diameter of 1 mm. After the selective wet chemical etching of the 1050°C thermal SiO₂ passivation layer, ohmic metal of Ni/Ti/Al/Au (35/50/100/100 nm) are deposited on the side of the detector and annealed by rapid thermal annealing in N₂. Finally, a thick pad metal of Ti/Au (500/1000 nm) is deposited for the subsequent wire bonding.

III. ELECTRON BEAM RESPONSE CHARACTERIZATION

Fig. 1c presents the temperature dependent I - V curves of the fabricated SiC detector with p-i-n structure, where dark current of 0.05 pA at -100V bias with superior stability at 200 °C could be observed. The C - V characteristics in Fig. 1d unveils that the

detector has a low background carrier density of $1.85 \times 10^{14} \text{ cm}^{-2}$, while over $5 \times$ reduction in capacitance is obtained with the thick-layer depletion. Fig. 2a and b shows the simulated energy deposition depth distribution of the electrons with various energies, where energy over 5 keV is expected obtain preferable depth across the current spreading layer. For the available electron beam current in the scanning electron microscopy (SEM), i.e. 0.78 pA to 410 nA, the detector output current shows evident dependence on the electron beam current, and maintains a contrast ratio over 10^4 versus the dark current, as shown in Fig. 2c. In Fig. 2d, evident response to electrons with energy over 5 keV is realized, especially for those over 15 keV, agreeing well with the simulation in Fig. 2b.

As presented in Figs. 3a and b, the detector has excellent response uniformity with scanning area varying from mm^2 to μm^2 , covering nearly the entire detector. In Fig. 3c, superior linearity could be observed between the detector current and the electron beam current, with a large dynamic over 10^4 . By taken the average electron-hole pair-creation energy as 7.8 eV for 4H-SiC [6], high electron detection efficiency of 95% is achieved with consideration of the energy deposition in the top 300-nm p-type layer, in Fig. 3d.

Detector reliability against the electron bombardment has been systematically evaluated, utilizing extremely harsh irradiation conditions of 1.86 GeV $^{181}\text{Ta}^{31+}$ and 1.30 GeV $^{84}\text{Kr}^{18+}$ heavy ions with total dose of 1×10^7 ions/ cm^2 , and γ -Ray of 7×10^4 Gy, as shown in Fig. 4a. Minor degradation could be observed in the sub-pA leakage in Fig. 4b. And, highly focused 30 keV electron beam in nm-scale with current density over 10^7 A/ cm^2 for 1h is applied. As presented in Fig. 4c, both the leakage and response current to 410-nA and 0.1-nA electron beam exhibit superior reliability, which should be attributed to the exceptional thermal conductivity, high temperature tolerance, and radiation hardness of the SiC detector.

IV. ELECTRON MICROSCOPY EXPERIMENTS

A. Scanning electron microscopy based on SiC detector

Fig. 5 illustrates the flowchart for morphology imaging based on the SiC electron detector. As the electron scans the sample surface line by line, the detector current is collected and fed into the transimpedance amplifier (TIA), where temporal voltage signals are generated, with voltage evolution directly relates to the surface features. Nearly periodical voltage signal in time domain could be obtained in the line-by-line scanning mode, based on which two-dimensional image could be reconstructed with precise identification of the boundary between each line. Fig. 6a presents the results as captured by the SEM with secondary electron imaging mode, while Fig. 6b and c present the time-domain voltage signals and the reconstructed image based on the SiC detector, respectively. High-degree reproduction of the image has been obtained, while the scanning electron image based on the SiC detector shows more shadowing effect for three-dimensional view, and also provides more compositional information, as highlighted at the positions of A and B. Figs. 6d-f gives a zoomed-in view of an particle at position A in Fig. 6c, where sub-micro meter structure features could be identified, achieving a spatial resolution of ~ 220 nm with gray intensity contrast of 10%-90%, and 34 nm for a visible gray intensity contrast of 10%-20%, as shown in Fig. 7a.

B. Single-electron-detection-based scanning microscopy

Single electron detection capability of the SiC detector is demonstrated by integrating dedicated front-end electronics,

including an active reset charge sensitive amplifier (CSA) logic [6] and a small reset capacitor of 100 fF for the first stage amplification, time to digital converter (TDC) based on vernier voltage-controlled ring oscillator for the electron incident time analysis [8], and analog to digital converter (ADC) for the electron-induced-charge and thus the electron energy analysis, as shown in Fig. 8a. Temporal voltage signal is firstly output from the CSA, which consists of the charge integration stage, and the reset stage to maintain the quasi-static operating point of the input stage of the amplifier. The incidence of electron would induce transient charge accumulation and thus a small voltage variation stage, which would then be filtered to obtain the charge-number-dependent voltage, as illustrated in Fig. 8b. Voltage pulses with height linearly related to the incident electron energy are obtained, revealing an energy conversion factor of $\sim 10 \text{ mV/keV}$. In Fig. 8c and d, with TDC analysis on the rise edge of the pulse, a standard error for the time analysis as low as 6.62 ns is obtained, which should be attributed to the superior time resolution of the TDC (92.5 ps) and the low capacitance of the detector with thick depletion layer at the operation bias of -100V.

With electrons scanning across an Au film as placed on a Al stage, temporal voltage evolution containing characteristic single electron signals from Au and Al could be obtained, as shown in Fig. 8e. The output voltage falls faster when scanning on the Au, which should be a result of the higher elastic electron scattering possibility for material with larger atom number, where inelastic-scattering-related extra energy loss is lower [9]. As shown in Fig. 8f, the different evolution stages could be converted into similar waveforms as in Fig. 4c, and thus obtaining spatial information. With statistical amplitude analysis applied on the separate pulses (Fig. 8g) with the slow shaper and ADC, single scattering electron spectroscopy is realized, as presented in Fig. 8h, evidencing the capability in single-electron-based topographic and compositional microscopy analysis.

V. CONCLUSION

As presented in the benchmark, this work demonstrates for the first time single-electron detection for temporal compositional and spatial microscopy featuring. With the well-modulated charge collection electric field and leakage suppression processing, the SiC detector shows wide dynamic range, high detection efficiency, and superior radiation hardness. While the direct decoding of the single electron energy and time information based on the dedicated electronics would facilitate the opening of new application dimensions in advanced IC nodes, quantum materials, biomolecular systems, as well as the astrophysical material studies.

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Working Principle and Basic Characteristics of the Device

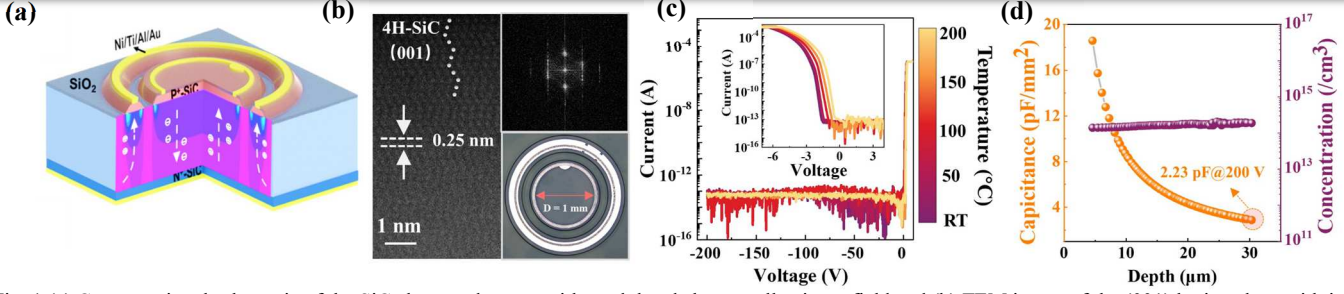


Fig. 1 (a) Cross-sectional schematic of the SiC electron detector with modulated charge collection e-field and (b) TEM image of the (001) lattice plane with insets shows the FFT diffraction pattern and the detector photograph. (c) Temperature-dependent dark current and (d) capacitance-voltage characteristics of the detector.

Electron Beam Response Simulation and Characterization

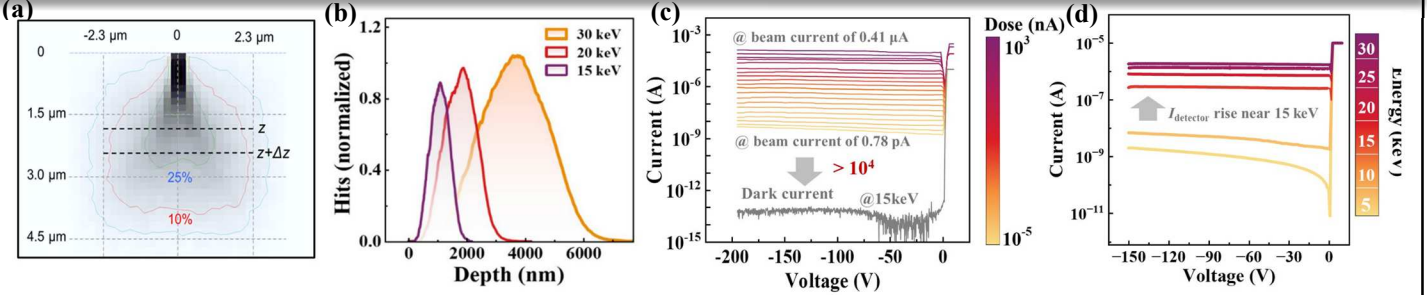


Fig. 2. (a) Profile and (b) longitudinal distribution of electron hits probability and thus the energy deposition for the incident electron with different energies in the 4H-SiC from Monte Carlo simulation. (c) Detector response current at various incident beam current and (d) different electron energies.

Electron Beam Response Characteristics

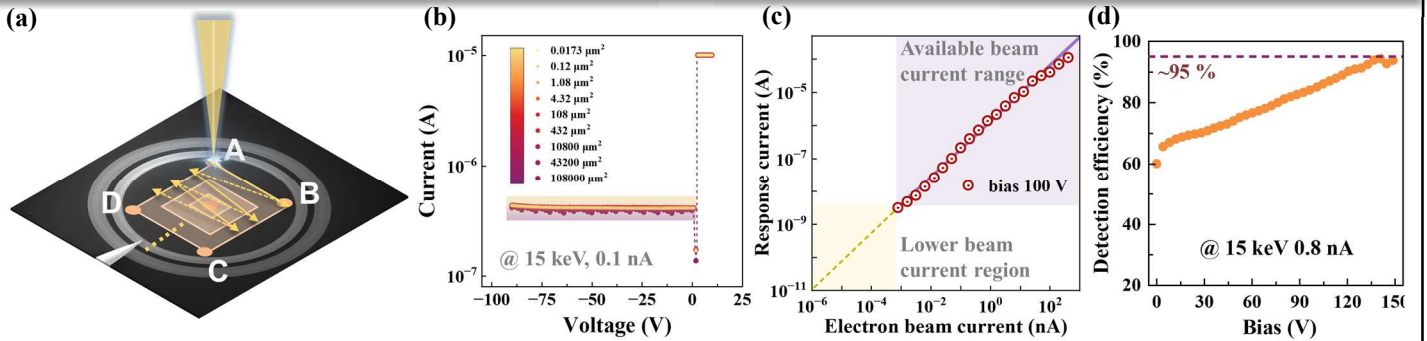


Fig. 3. (a) Schematic illustration of electron response uniformity evaluation setup with different scanning areas and (b) the corresponding detector current results. (c) Linearity evaluation results in the dynamic range with beam current varying from 0.78 pA to 410 nA. (d) Detection efficiency evaluation with the average electron-hole pair-creation energy taken as 7.8 eV, evaluated at 15 keV and a moderate beam current of 0.8 nA, with energy deposition in the top layer

Reliability and Radiation Hardness Evaluation

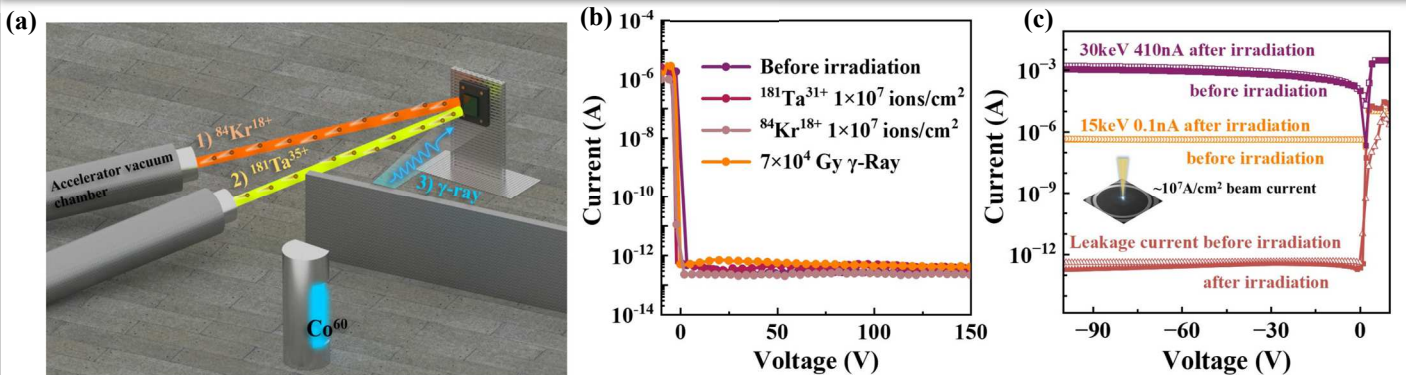
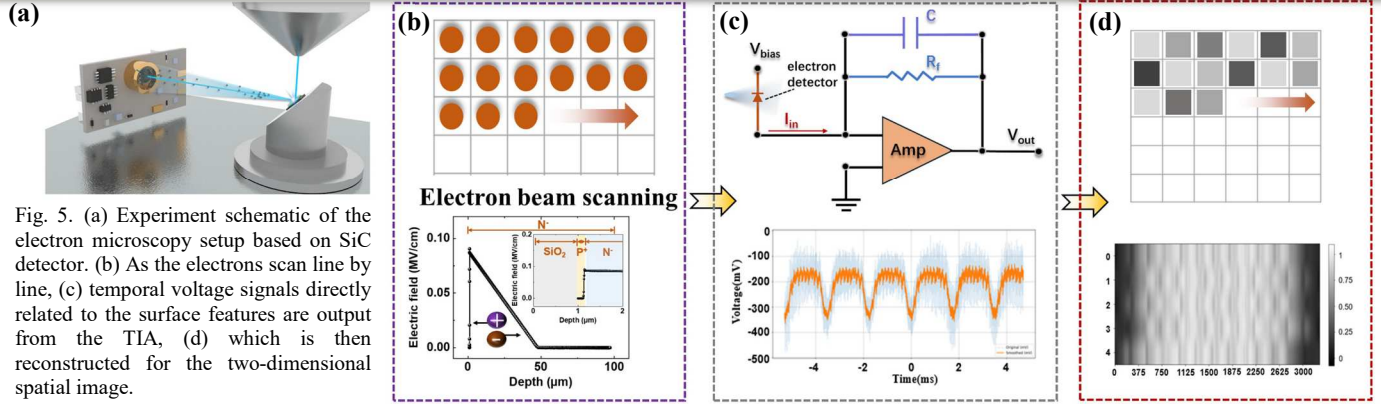


Fig. 4. (a) Schematic diagram of the radiation hardness evaluation experiment with 1.86 GeV $^{181}\text{Ta}^{31+}$ and 1.30 GeV $^{84}\text{Kr}^{18+}$ heavy ions, and γ -Ray irradiations. (b) Detector leakage characteristics before and after the harsh irradiations. (c) Detector leakage and response current characteristics before and after the 30-keV electron beam irradiation with density of $\sim 1.02 \times 10^7$ A/cm² for 1 hour.

Electron Microscopy Experiment Flowchart



Electron Microscopy Characteristics

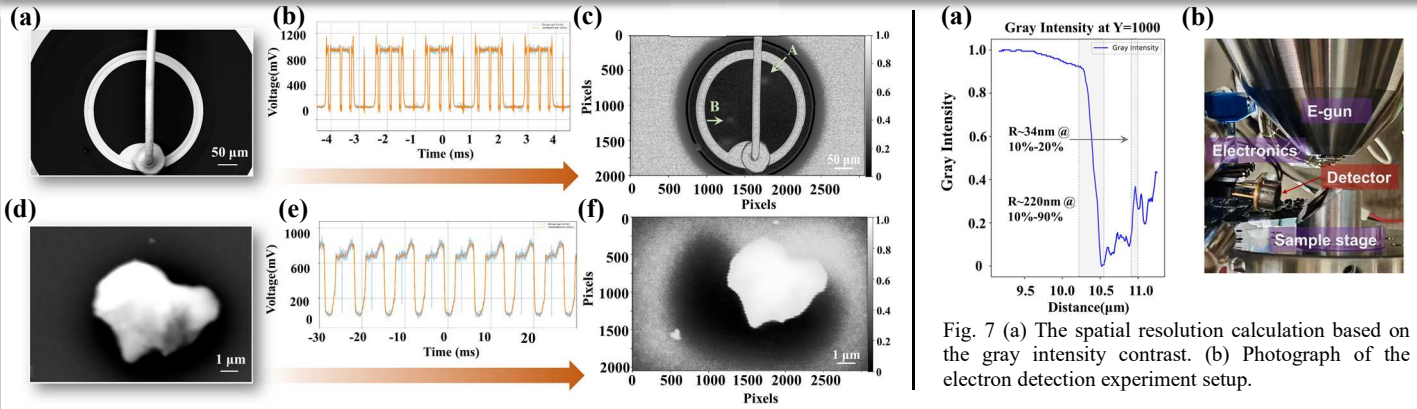
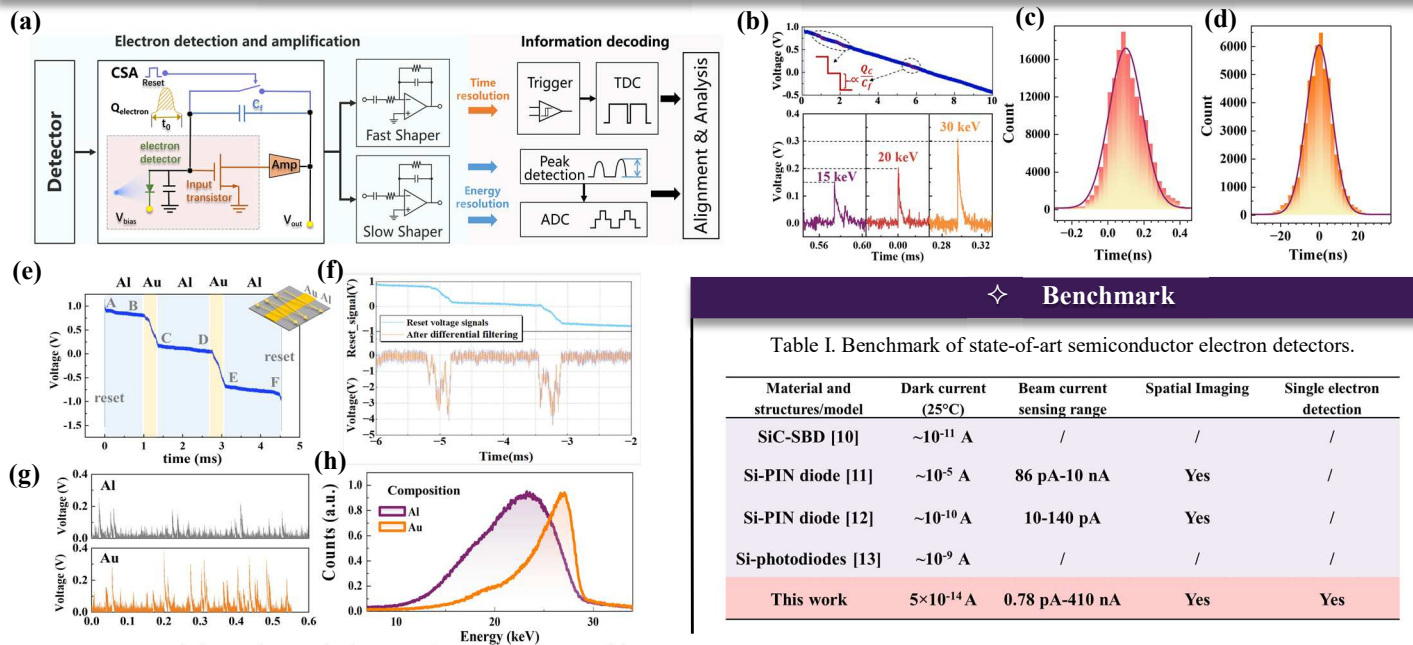


Fig. 7 (a) The spatial resolution calculation based on the gray intensity contrast. (b) Photograph of the electron detection experiment setup.

Single-electron energy spectroscopy and real-time compositional microscopy



Benchmark

Table I. Benchmark of state-of-art semiconductor electron detectors.

Material and structures/model	Dark current (25°C)	Beam current sensing range	Spatial Imaging	Single electron detection
SiC-SBD [10]	$\sim 10^{-11}$ A	/	/	/
Si-PIN diode [11]	$\sim 10^{-5}$ A	86 pA-10 nA	Yes	/
Si-PIN diode [12]	$\sim 10^{-10}$ A	10-140 pA	Yes	/
Si-photodiodes [13]	$\sim 10^{-9}$ A	/	/	/
This work	5×10^{-14} A	0.78 pA-410 nA	Yes	Yes